Below the Noise Floor: A Deterministic GLEE Agent and the Measurement Problem It Exposed

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Abstract

I study the difficulty of evaluating small strategy changes in a live economic-game competition through a deterministic, LLM-free agent. The agent placed sixth in the GLEE agent track; its development fleet logged 264,432 games across bargaining, negotiation and persuasion. In a simultaneous 8.6-hour comparison, five identical policy builds accumulated approximately 3,000 games each yet differed by as much as 88 overall rating points and 293 within a game family. This documents substantial leaderboard dispersion without a policy change, while leaving the contributions of initialization, matchmaking and a changing reference pool unresolved. Decision-level replay and terminal acceptance counterfactuals help investigate the behavior behind these scores: some interventions changed no actions, while transcript inspection exposed message-parser and termination-limit failures. An observational comparison with my second-place human-track play shows different participation behavior, but does not isolate message-understanding ability. The results motivate simultaneous controls, explicit rating-model assumptions and behavioral audits when evaluating small interventions during short competition windows. Code and analysis scripts are public; full empirical reproduction depends on the underlying game records.

1 Introduction

GLEE [10] evaluates strategic interaction in bargaining, negotiation and persuasion. Its competition provides a setting in which to ask a practical question: *how can an entrant distinguish a useful policy change from a noisy score change?* I address this through 7AIDARA_GAMMA, a deterministic agent that finished sixth with an overall rating of 2360.8, and through my own human-track play, which finished second.

The contribution is an empirical case study of evaluation and behavior rather than a new equilibrium solution. Automated-negotiation research already studies agent evaluation, modular strategies and acceptance conditions [1, 3, 2]; uncertainty in benchmark comparisons is also an established concern [11]. Here, five simultaneous identical-policy controls reveal rating dispersion in a live competition, and offline audits distinguish changed behavior from changes that never affected an action. Failed experiments and implementation errors are part of the evidence, not merely development history.

2 The agent and its operating assumptions

A dispatcher routes games to three Python policies (858 / 660 / 781 lines), configured by 114 tunables. Parameters are fixed during a run; within-game state and belief updates are deterministic given the observed history. Seller randomization is seeded from the game ID and round. A safety layer repairs

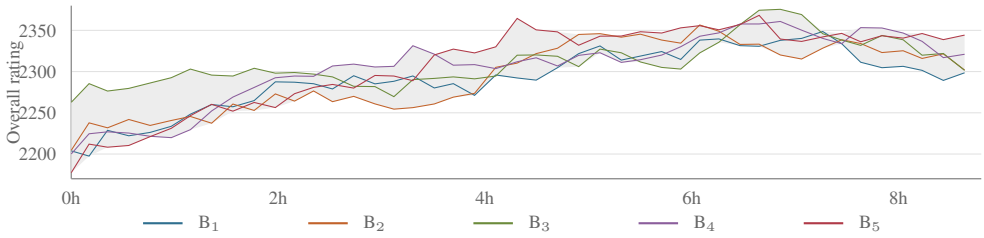


Figure 1: Overall ratings of five identical policy builds over 8.6 hours. Shading is the observed min–max envelope, not an uncertainty interval. The mean spread is 48.1 points and its maximum is 88.0. Initial rating states were not reset.

invalid actions and supplies a legal fallback after an exception. The reference build is covered by 472 behavioral tests.[1]

Bargaining. The policy uses the infinite-horizon proposer-share anchor $(1 - \delta_{\text{them}})/(1 - \delta_{\text{us}}\delta_{\text{them}})$ [9], with empirical floors, concession rules and explicit endgame handling. This is a heuristic benchmark, not a best-response guarantee against arbitrary opponents. Finite-horizon equilibrium predictions depend on the remaining horizon and seat; the agent instead exploits the empirically favorable final-proposer position (91.9% agreement in a 1,415-game cohort), while accounting for termination risk.

Negotiation. Observed values came from $(80, 100, 120, 150)$ scaled by $\{1, 10^2, 10^4\}$, with both players on the same scale. The policy therefore prices at possible tradable rungs. Independent uniform draws on that ladder predict positive surplus in $6/16 = 37.5\%$ of value pairs; independence of roles alone would not imply that rate. When observed information rules out profitable trade, the policy walks away.

Persuasion. Consistent with GLEE's documented payoff structure, the competition seller received the full price on a sale regardless of quality (checked over 1,829 sold rounds). For a buyer who trusts a known recommendation policy and purchases at posterior threshold τ, recommending every high unit and a fraction q of low ones gives $\Pr(H \mid \text{recommend}) = p/[p + (1-p)q]$ and $q_{\max} = \min\{1, p(1-\tau)/[\tau(1-p)]\}$. Because GLEE lacks commitment, this Bayesian-persuasion threshold [6] is a heuristic starting point, not a guarantee of optimality. The seller ramps its budget with the horizon. The buyer combines message parsing with within-game purchase outcomes and recommendation frequency; quality is hidden after a decline.

3 Identical policies, different leaderboard histories

Five byte-identical builds were polled together for 8.6 hours, producing 45 simultaneous snapshots over approximately 3,000 games per agent. The overall min–max spread averaged 48.1 points and reached 88.0; the largest within-family spread was 292.9 (Figure 1). These are descriptive spreads, not confidence intervals. The five agents had different initial ratings and match histories, and successive polls are dependent. The experiment measures their joint effect with sampling and pool changes, rather than isolating one source of noise.

Competition scoring compares payoffs within a configuration and role and adjusts for opponent strength. For interpreting late-stage updates, I used the simplified EMA model $R_{t+1} = (1-\eta)R_t + \eta g_t$, with $\eta \approx 0.002$ inferred from rating deltas and g_t denoting an adjusted game target. At constant η, its half-life is $\log(0.5)/\log(1-\eta) \approx 346$ family games. Under independent, stationary targets, its effective sample size is $(2-\eta)/\eta = 999$; lifetime game volume therefore need not describe the information in the current rating. These are model implications, not a reconstruction of every server update.

The reference pool changes as games enter it, making unmatched cross-time comparisons confounded. Selecting or freezing a high-scoring slot also makes stopping time part of the evaluation protocol. A further distinction concerns the outcome being tested: for consecutive family updates,

[1] **LLM usage.** The competition agent calls no language model. I used LLM coding assistance for analysis scripts and for drafting and revising the paper.

$\overline{\Delta R} = (R_{\text{end}} - R_{\text{start}})/n$. Mean rating change therefore tends toward zero as a rating settles, even for a strong policy. An IID calculation based on such deltas cannot by itself establish a sample requirement for detecting a difference in policy strength (Appendix D).

4 Behavioral measurements beyond the displayed rating

Terminal acceptance counterfactuals. An available offer can be evaluated exactly at its decision point: accepting banks $\text{share}\,\delta^{r-1}$. Comparing that value with the realized continuation made 23,617 refusals informative about acceptance rules, including continuations ending without agreement. This is a local, retrospective comparison; it does not identify the value of every alternative policy or prove ex-ante optimality. Repeated refusals within one game share an outcome.

Decision-level replay. The reference policy reproduced 4,272 of 4,272 bargaining and 13,056 of 13,056 negotiation decisions in the audited cohorts. Replay then counted actions changed by a candidate flag, rather than how often its condition held. One condition fired on 43% of eligible games yet changed 0 of 13,056 decisions; another changed 0 of 13,479. Exact replay validates the implementation on logged states, not the outcome of unobserved trajectories.

Exploratory break-even fits. Regressing rating delta on banked payoff within a cell supplied diagnostic zero crossings (Appendix Table A3). Slopes were positive in 58 of 64 cells, with median $R^2 = 0.60$ and split-half correlation $r = 0.741$. These describe regularity in the logged sample. A root depends on rating history, opponent adjustments and the reference pool; it is not generally a recovered field median. A fourth diagnostic estimates aggregate quality among declined persuasion rounds from the known prior (Appendix B).

5 Interventions and implementation failures

Across 30+ interventions and 49 runs, the clearest improvements concerned implementation repairs and structural policy changes (Appendix Table A4). Their evidence types differ; none of the empirical comparisons alone establishes game-theoretic dominance. Seven consecutive tuning arms yielded no reliable improvement in the comparisons used. A seller lie-rate arm also showed no reliable difference across three runs ($t = -0.29$), which does not establish equivalence.

Transcript inspection found purchases following messages such as "Not this one… pass this round." The parser recognized "not this round" but not "not this one" and interpreted unmatched text as a recommendation. Re-scoring 123,700 buyer rounds reclassified 270 purchases, 258 of them low quality. Another transcript exposed an acceptance gate opening at round 97 in negotiations that terminated at 96. A separate message intervention changed take-up from 0.689 to 0.693; that observation does not establish that language generally has no effect.

On the final morning, a fleet-wide concession change was followed by drops of 100–385 negotiation rating points over five hours. Without a concurrent untreated control, the full decline cannot be attributed to the parameter alone. Operationally, the lesson is to retain a control. Abandonment was also clustered: 80 of 109,881 rated games, with 78 in seven clock hours. Their recorded deltas averaged -6.76 and summed to about -541; the sum is not their persistent effect on an EMA rating.

6 Agent Behavior Analysis

Observed behavior. Bargaining settled at median round 2, with 1.5% no-deals and mean banked share 0.429. Negotiation walked away in 79.8% of untradable seats and agreed in 20–24% of hidden-value games versus 85–89% with complete information. The persuasion buyer bought 17.4 units per game where buying blind had positive expected value, versus 2.2 where it did not. These are sample descriptions, not controlled treatment effects.

Design alignment and failures. The 472 tests encode intended actions in named regimes. The reference snapshot passed them; the final-night working tree, carrying experimental flags, failed 27. Together with replay, these tests check implementation consistency rather than utility optimality. Four recurring deviations were unreachable rules, a gate keyed to the wrong termination limit, a permissive parser default and excessive last-round concession. In one bounded negotiation cohort, 403 of 469

Table 1: Observational human–agent comparison. Bargaining: mean banked share. Persuasion: payoff per game in product-price units; t has Welch's standard-error form, signed human minus agent, without dependence adjustment. Persuasion rows contain 48–92 human and 4,729–9,467 agent games.

Bargaining (known horizon)				**Persuasion (buyer)**			
seat	cells	human	agent	regime	human	agent	t
player 1	14	0.444	0.447	$mp > 1$	$+24.22$	$+22.10$	$+1.31$
player 2	13	0.512	0.720	$mp = 1$	$+2.99$	$+2.56$	$+0.56$
				$mp < 1$	-0.61	$+0.18$	-2.44

seller games reached a round-10 agreement at low surplus (Appendix Table A5). Agreement rate alone concealed that behavior because capitulation still counted as a deal.

7 An observational same-author human baseline

I played 2,690 rated human-track games and finished second. Sharing the game families and opponent pool makes a useful case study, but does not match exact opponents, messages or times. Table 1 compares shared bargaining cells and pooled persuasion regimes. Bargaining equally weights retained cells matched on horizon type, seat and discount factors; persuasion pools by mp, so within-regime configuration composition may still differ.

Among purchased items, the human's high-quality rate was 0.738 over 2,745 purchases and the agent's was 0.725 over 59,603. These rates concern selected purchases, not message-reading accuracy. The agent's higher bargaining share in the known-horizon second seat, and my negative mean persuasion payoff where $mp < 1$, are consistent with different participation and acceptance behavior. They do not identify discipline rather than perception as a causal mechanism, nor support a general claim about humans versus machines.

8 Limitations and conclusion

This is one account, one human participant and a changing competition pool. The identical-policy experiment does not reset rating history or isolate matchmaking; exploratory fits and pooled tests do not remove all dependence or selection. Repeated tuning and flexible cell filters increase the risk of selective findings. The public repository provides policies, tests and analysis scripts, but full empirical reproduction requires private game records and deployment metadata; anticipated organizer data do not by themselves resolve that gap.

The useful finding is narrower than a universal failure of leaderboard evaluation: substantial rating dispersion occurred without policy changes, and behavioral audits exposed failures that score summaries alone obscured. Simultaneous controls, explicit estimands, replay and transcript inspection provide complementary evidence when live experimental windows are short.

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A Supplementary behavioral evidence

Table A1: Buyer-template history and high-quality rate on the next purchase (source cohort: 6,185 buyer replays). Rates are conditioned on purchases and template history; cohort size is not the event count in each row. The all-low three-times estimate is below every configured immediate break-even threshold $1/m$, motivating a veto heuristic rather than proving dominance.

Prior record of template	hit rate	Prior record	hit rate
no prior	70.4%	all-low $\times 1$	33.8%
all-high $\times 1$	81.3%	all-low $\times 2$	19.3%
all-high $\times 2$	84.8%	all-low $\times 3$	5.3%
all-high $\times 3$	91.2%	mixed	63.7%

Table A2: Development-fleet volume, 12–30 August 2026, recovered from local turn logs. These are fleet totals, not the games of the single ranked slot.

Family	unique games	Fleet total	
Bargaining	89,641	logged turns	3,114,356
Negotiation	76,490	unique games	264,432
Persuasion	98,301	reconstructable runs	49

Table A3: Exploratory break-even diagnostics from 7,104 known-horizon and 31,208 open-horizon bargaining games. A fitted zero crossing is conditional on the observed rating process; it is not generally a field median. A separate 26,773-decision replay cohort had 98.79% control fidelity and exposed a gate as low as 0.118 in the known-horizon player-1 regime. It is distinct from the exact-replay cohorts reported in the main text.

Horizon	Seat	Mean banked	Fitted zero crossing	Difference
open	p_1	0.530	0.476	$+0.054$
open	p_2	0.501	0.463	$+0.038$
known 12	p_1	0.435	0.816	-0.381
known 12	p_2	0.687	0.449	$+0.239$

Table A4: Selected policy changes and their diagnostic evidence. Evidence types and cohorts differ: live comparisons, local replay and transcript reclassification do not establish a common replication standard or game-theoretic dominance. Standardized statistics are reported as exploratory effect-to-noise summaries.

Change	Evidence
Complete-info ultimatum	Larger demand in a one-round known-surplus seat; reported $+3.42\sigma$, $n = 214$. Acceptance remains an opponent decision.
Rung-aware pricing	Two reported live comparisons: $+3.9\sigma$ and $+4.1\sigma$.
Free-clock accept floor	Observed gain $+0.0458$ of pot in 108 games, reported 7.7σ, with none worse in that cohort. No discounting does not eliminate termination or rejection risk.
Open-horizon cap response	Replay identified 11 of 11 examined zero-payoff games affected by the termination-limit failure.
Negative-signal parser	270 purchases reclassified; 258 (95.6%) low quality, versus a 27.5% overall purchase-conditioned low-quality rate.

Table A5: Selected complete-information seller cells from a 10,740-game negotiation analysis. The displayed seller cohorts contain 957 games; the 469-game horizon-10 cohort is partitioned below. The association between late settlement and low payoff does not identify the causal effect of waiting.

Cell / outcome	n	Payoff measure	rating / game
$\mathrm{max_rounds} = 1$	488	asks 0.991 of surplus; 99.2% sign	$+6.10$
$\mathrm{max_rounds} = 10$	469	partitioned below	-4.63
closed by round 8	66	banked / buyer value: 0.2365	$+5.9$
agreed in round 10	403	banked / buyer value: 0.0433	-6.4

B Additional methods and limits

Aggregate inference for declined products. Let p be the prior probability of high quality in a persuasion cell, b the purchased fraction, and h the observed high-quality rate among purchases. The population identity $p = bh + (1-b)q_{\mathrm{declined}}$ motivates

$$\widehat{q}_{\mathrm{declined}} = \frac{p - \widehat{b}\,\widehat{h}}{1 - \widehat{b}}, \qquad \widehat{b} < 1.$$

This uses the prior in expectation, not the realized total number of high-quality units in a finite sample. It estimates an aggregate rate and cannot recover the quality of a particular declined product. Near-complete purchasing makes the estimate unstable; sampling variation can place it outside $[0, 1]$. At $b = 0$ it supplies no information beyond the prior. It should not be used as evidence about a single hidden round or as a precision estimate without uncertainty.

Simulation and memory. A local bargaining simulator fitted to 15,788 games and eight adversarial archetypes reproduced per-cell median settlement rounds but overestimated payoff by approximately 0.03 of the pot. It was used for candidate screening rather than as evidence of a live treatment effect. Opponent profiles showed split-half correlation $r = 0.786$, but only 19 opponent-by-cell profiles had at least 25 purchases in each half. Identity was withheld in 49.9% of games, and no quality was revealed in 42,616 unbought rounds. Purchase selection and sparse profiles limit the conclusions for cross-game opponent modeling [4].

Operational versioning. Experimental flags written into a working tree, entry points outside a copied package and abrupt mid-game termination can make intended and deployed behavior differ. A reference build passing tests does not establish that every concurrent slot ran that build. Exact replay is tied to the logged cohort and policy version.

C Aggregation and outcome-dependent selection

A candidate rule used whether a seller had previously declined an opportunity to sell. Pooled across fifteen (m, p) cells, later low-quality rates were 0.285 / 0.287 / 0.286 / 0.315 for sellers declining 0 / 1 / 2 / 3–4 of their first four opportunities. Within $(m = 1.20, p = 0.80)$, the corresponding values were 0.160 / 0.102 / 0.081 / 0.069 (cell sample $n = 9,932$), against an immediate break-even low-quality rate $1 - 1/m = 0.167$. This illustrates how configuration mixture can mask a within-cell pattern; it is not evidence of a uniform reversal in every cell. The rule was not deployed.

Separate analyses were affected by outcome-dependent inclusion: deal rates computed only among closed deals, quality estimates conditional on purchases, and abandonment estimates from logs missing a completed final turn. These are selection problems, distinct from aggregation over cells. One persuasion arm also changed from -3.11σ to -0.82σ to $+3.02\sigma$ when the minimum-count filter admitted 3, 33 or 44 cells. This sensitivity motivates specifying inclusion rules before inspecting outcomes and treating retrospective screening statistics as exploratory.

D Rating-model and reproducibility boundaries

An idealized independent two-arm comparison with equal variance σ^2 and equal sample size n has standard error $\sigma\sqrt{2/n}$. Requiring an effect Δ to equal three standard errors gives $n = 18\sigma^2/\Delta^2$.

Earlier screening applied this arithmetic to per-game rating deltas, whose observed standard deviations were 4.84 / 4.42 / 4.44 over 120,023 joined games. At $\Delta = 0.25$, it yields roughly 5,600–6,700 observations per arm using unrounded estimates. This is an IID sensitivity illustration, not a validated policy-strength detection bound: deltas depend on current ratings and telescope over a run, and the criterion does not specify statistical power. Fixed-reference game scores or properly reconstructed game targets, with time and configuration controls, would provide a better-defined comparison.

Similarly, a fitted zero rating-delta crossing is not generally the median of the current payoff distribution. Rating history, opponent strength and a changing reference pool enter that relationship. Comparisons with equilibrium predictions additionally require the correct initial seat and horizon; an infinite-horizon proposer share must not be treated as the equilibrium payoff for all seats and finite horizons.

For a constant-η EMA, perturbing one game target by d changes the rating by $\eta d(1-\eta)^k$ after k later games if those later targets remain fixed. The recorded sum of abandonment deltas therefore cannot be interpreted as a final-standing effect. A persistent stream of abandonments and a finite cluster near stopping time are different estimands; no precise persistent-cost estimate is claimed here.

The public release includes policy code, tests and analysis scripts. Its scripts require joined game records and human-history exports that are not distributed alongside the code because they include opponents' transcripts. Organizers have indicated plans to release game records, but local deployment flags, rating-poll snapshots, filter choices and joins are additional provenance requirements. The released method is inspectable; the empirical results are not yet a one-command reproduction. Tables retain their analysis-specific cohorts, and aggregate counts should not be substituted for the denominators of individual comparisons.